Monitoring Global Gender Inequality and Child Labor using Facebook

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1. Abstract

The increasing need to address global gender inequality and child labor has prompted a data-driven approach to understanding and mitigating these issues. This project explores the interconnections between gender disparities and child labor across countries, leveraging comprehensive datasets from the World Bank, International Labour Organization (ILO), UNICEF, and Facebook API. The primary objectives were to evaluate and optimize an existing Global Gender Gap Index (GGI) model, develop a predictive model for child labor, and establish the correlation between these phenomena.

Our methodology integrates innovative modeling techniques, including Elastic Net regression and feature engineering, to predict GGI and child labor trends. The project employed multiple indicators, such as economic participation, educational attainment, political empowerment, and health metrics, to construct robust predictive frameworks. Notably, we incorporated the Facebook Gender Gap Index (FBGGI) as a supplementary variable to enhance prediction accuracy, despite the challenges of overfitting in some models.

Key findings highlight the predictive efficacy of Elastic Net, achieving R^2 values as high as 97.54% for GGI predictions. Similarly, our child labor model underscored significant socio-economic factors, such as the Human Development Index (HDI) and labor force participation rates, as critical indicators. An interactive dashboard was developed to visualize these findings, offering stakeholders actionable insights.

Despite challenges like data sparsity and the complexity of hyperparameter tuning, the study provides a comprehensive framework for understanding gender disparities and child labor. Future work will focus on integrating time-series data and deepening the analysis of tertiary education's role in mitigating these issues, aiming to develop sustainable solutions for global inequalities.

2. Introduction

Gender inequality and child labor remain pervasive issues in many parts of the world, particularly in developing countries. Addressing these challenges is critical for achieving sustainable development goals (SDGs) and fostering equitable societal progress. The intersection of gender disparities and child labor, although intuitively related, is often underexplored in existing research. This report investigates the potential correlation between gender inequality, measured through innovative online metrics, and the prevalence of child labor across different nations.

The Global Gender Gap Index (GGI), published annually by the World Economic Forum, serves as a benchmark for assessing gender disparities. However, its annual cadence limits the ability to detect real-time sociodemographic shocks or monitor progress at higher frequencies. To address this limitation, the Facebook Gender Gap Index (FBGGI) is proposed as a supplementary metric, leveraging digital footprints to measure online gender disparities. FBGGI, defined as the ratio of the female-to-male gender ratio of Facebook users to that of

the general population, provides a dynamic and timely indicator of gender gaps in the digital space, often mirroring offline inequalities.
Simultaneously, child labor statistics reveal that in the least developed countries, over 20% of children aged 5 to 17 are engaged in labor detrimental to their health and development. Gender-specific dynamics are suspected to play a role in this alarming trend, suggesting that disparities in gender equality may influence or exacerbate child labor practices. This study hypothesizes that online gender inequality, as captured by FBGGI, and traditional offline gender disparities are linked to the prevalence of child labor.
Using a quantitative framework, we establish models to examine these relationships:
The foundation of our models is the following formula for predicting the Global Gender Gap Index (GGI) and child labor:

$$GGI = w_1 \times FBGGI + w_2 \times Indicator_1 + w_3 \times Indicator_2 + \cdots + w_n \times Indicator_n$$

$$Child\ Labor = w_1 \times FBGGI + w_2 \times Indicator_1 + w_3 \times Indicator_2 + \cdots + w_n \times Indicator_n$$

Here, FBGGI represents the Facebook Gender Gap Index, a key predictor variable, while the Indicators are various socio-economic and demographic factors sourced from organizations like the World Bank and ILO. The weights (w_i) are optimized through model training to best capture the influence of these indicators on the target variables.

To achieve our objectives, the project followed a structured approach:

1. **Data Acquisition:** Comprehensive data was collected from multiple sources, including the World Bank, UNICEF, and Facebook API, ensuring a robust dataset for modeling.
2. **Modeling Techniques:** Elastic Net regression was employed to predict GGI and child labor trends, balancing complexity and generalizability.
3. **Feature Engineering:** Indicators were carefully selected and engineered to improve model performance, with an emphasis on reducing multicollinearity and overfitting.

By incorporating various-level variables alongside FB GGI and other relevant data, this report aims to establish a comprehensive understanding of the interplay between gender inequality and child labor. The findings could provide actionable insights for policymakers, enabling more frequent monitoring and targeted interventions to mitigate these challenges. In doing so, this study contributes to advancing SDGs by fostering equitable online and offline societies and protecting vulnerable populations from exploitative labor practices.

3. Dataset

3.1 Dataset Description

The dataset for this project was meticulously compiled from reputable sources such as the World Economic Forum, UNICEF, the World Bank, and the World Health Organization (WHO). Leveraging Pandas, data from these diverse sources were harmonized and merged to ensure consistency and accuracy. Following the project requirements set by our sponsors, MacroX Studio, the Gender Gap dataset focuses mainly on nine countries: Afghanistan, Bangladesh, Germany, India, Japan, Nigeria, Pakistan, Tanzania, and Zambia. The data spans four critical dimensions—economic opportunities, education, health, and political leadership—providing a comprehensive foundation for analyzing and comparing these aspects across the selected nations. **Figure 1** shows how the data appears for the 9 countries with a few indicators.

country	female_unemployment"%'	male_unemployment"% "	female_laborforce"% "	male_laborforce" %"	female_primaryeducation'	male_primaryeucation"%
Afghanisthan	27.288	13.483	6.53	60.289	51.5	53.12
Bangladesh	7.529	3.875	25.51	51.798	38.21	41.26
Germany	2.805	3.255	51.757	56.691	62.78	63.7
India	4.059	4.218	15.608	43.463	47.15	44.6
Japan	2.342	2.767	51.157	48.583	53.95	54.88
Nigeria	4.243	2.164	22.735	26.419	45.72	41.05
Pakistan	7.593	4.867	21.813	63.545	47.63	46.1
Tanzania	3.535	1.726	62.573	69.509	54.67	53.25
Zambia	6.22	5.664	31.385	38.316	54.73	52.55

Figure 1. Brief layout of the Gender Gap Dataset for the 9 countries

The first column 'Country' has the 9 countries and the remaining columns are all numerical values for various indicators such as unemployment rate, primary, secondary and tertiary education, life expectancy rate, mortality rate, the proportion of seats in politics, internet, mobile, and facebook gender gap data for both male and female.

The target variable for our child labor analysis came from UNICEF and was combined with world economic indicators sourced from the World Bank, NIAID, ILO, FRED, and the World Economic Forum. The dataset includes 39 features, encompassing socio-economic variables (e.g., GDP, education rates) and child labor indicators, providing a multi-dimensional perspective on the factors influencing child labor. Data was collected for 328 countries/ regions, representing a diverse range of economic and social conditions. Although the data spans five years, the target variable is a single value for a range of years, making a temporal analysis of child labor trends uninformative. Consequently, we focused solely on the year 2023 for this analysis. While the dataset is extensive, gaps exist across countries and years, requiring careful handling during preprocessing.

3.2 Data Preprocessing

Data preprocessing was a crucial step in preparing the datasets for further analysis. This process involves several stages, including merging multiple datasets, handling missing values, standardizing variable names, and ensuring consistency across features. We focused on cleaning and transforming the data to ensure accuracy and remove any discrepancies, particularly with the country names, which were inconsistent across different sources. Additionally, we addressed outliers and potential biases in the data to enhance the robustness of the analysis.

Unlike the child labor dataset, the Global Gender Gap Index (GGI) dataset required minimal preprocessing due to its completeness and well-organized structure. The dataset was sourced from reliable databases such as the World Economic Forum and supplementary sources, with no missing values or significant inconsistencies in the data. All variable names were already standardized, and the data was consistently formatted across years and countries.

The child labor dataset required more intensive preprocessing due to its 39 features associated with the target variable. The first step involved merging data from UNICEF, the World Bank, and NIAID. A key challenge during this process was reconciling inconsistent naming conventions for countries across the datasets. Once the country names were standardized, the data was accurately merged. Next, missing values in the combined dataset were addressed. Countries with more than 50% missing data were excluded, reducing the dataset to 190 countries. To handle the remaining missing values, a combination of imputation techniques was employed. Linear regression was used for features with clear linear patterns specific to each country. For other missing values, a KNN imputer with $k = 3$ was applied, averaging the values of the three closest country-year combinations to ensure accurate imputations. After completing the imputation process, the years 2019–2022 were removed from the dataset, as the target variable was constant over a range of years. This decision ensured that a temporal analysis of child labor trends, which would have been uninformative, was avoided.

Feature selection is a critical step in building a predictive model, as highly correlated variables can be redundant and unnecessarily increase model complexity without adding significant predictive value. In our dataset, several features exhibited high correlations with others, prompting the removal of six variables: fertility rate, life expectancy, infant mortality rate, number of mobile cellular subscriptions, electricity access (%), and mean years of schooling. These features were strongly correlated with multiple other variables in the dataset, and their inclusion risked overfitting the model without providing meaningful performance improvements. By removing these redundant features, we ensured a more streamlined and efficient dataset for model training.

Data standardization and normalization are also essential steps in preparing data for predictive modeling, ensuring that all features are on a consistent scale. Various methods were tested, including standard scaling, log transformation, and power transformations, to determine the most effective approach. Yeo-Johnson power scaling emerged as the preferred method due to its ability to handle negative values, unlike the Box-Cox transformation. The Yeo-Johnson transformation uses maximum likelihood estimation to identify the optimal lambda parameter for data transformation. Depending on the lambda value and whether the input is positive or negative, different formulas are applied, allowing the method to effectively reduce skewness, inflate low variance, and deflate high variance to create more symmetric data distributions (C# Corner).

In conclusion, thorough data preprocessing was essential to prepare our datasets for analysis. This included merging data from various sources, handling missing values, selecting relevant features, and standardizing variables to ensure consistency and accuracy. These steps addressed inconsistencies, reduced redundancy, and enhanced the quality of the data for modeling. With the preprocessing complete, we are now ready to begin exploratory data analysis and develop predictive models to gain deeper insights into the factors influencing child labor and the gender gap index.

4. Exploratory Data Analysis

4.1 Gender Inequality EDA

The GGI dataset provides a comprehensive overview of gender-related indicators across nine diverse countries: Afghanistan, Bangladesh, Germany, India, Japan, Nigeria, Pakistan, Tanzania, and Zambia. It captures a multifaceted view of gender disparities through economic, educational, health, political, and social dimensions. Economic indicators include female and male unemployment rates alongside their respective labor force participation percentages, reflecting employment disparities. Educational data highlights the proportion of each gender's participation in primary, secondary, and tertiary education relative to the country's total population, unveiling gaps in access to education. Health metrics include life expectancy for females and males and suicide mortality rates, which offer insights into gendered health outcomes. Political representation is examined through the proportion of seats held by women and men in national parliaments, shedding light on decision-making power. Social dimensions are explored through measures like the Internet and Mobile Gender Gap, Facebook Gender Gap Index, Literacy Gender Gap, and various systemic and structural factors, such as Legal Gender Discrimination, Conservative Systems, and Occupational Segregation Gender Gap. Additionally, indicators like Physical Autonomy Gender Gap provide a nuanced understanding of personal freedoms. This rich dataset sets the stage for an in-depth analysis of gender inequality across countries with varying cultural, economic, and political contexts.

Building upon this comprehensive dataset, we will now delve deeper into each of these dimensions to explore the intricacies of gender inequality across these nine countries. By examining the economic, educational, health, political, and social indicators, we aim to uncover patterns, disparities, and correlations that highlight the unique challenges faced by each country.

4.1.1 Economic Participation and Opportunity

Economic participation and opportunity are key dimensions of gender equality, reflecting disparities in access to employment, workforce engagement, and earning potential. Globally, significant gaps persist between males and females in terms of unemployment rates, labor force participation, and the proportion of wage and salaried workers.

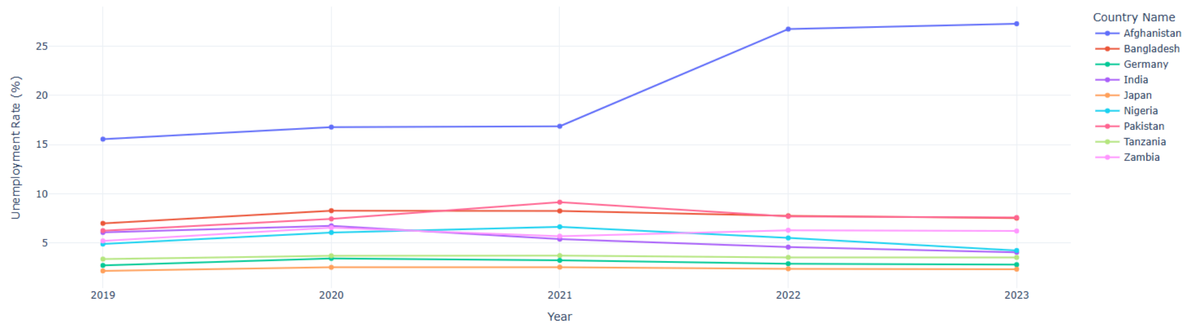

Figure 2. Female Unemployment Rate Trends by Country (2019 - 2023)

Figure 2. highlights female unemployment trends across nine countries from 2019 to 2023. Afghanistan records the highest rates, peaking at 26.75% in 2022, while Japan, Germany, and Tanzania maintain low, stable rates around 2%, 3.2%, and 3.5%. Pakistan's rate fluctuated, rising to 9.15% in 2021 before stabilizing at 7.71% in 2022 and 2023. These trends reflect varying economic dynamics across regions.

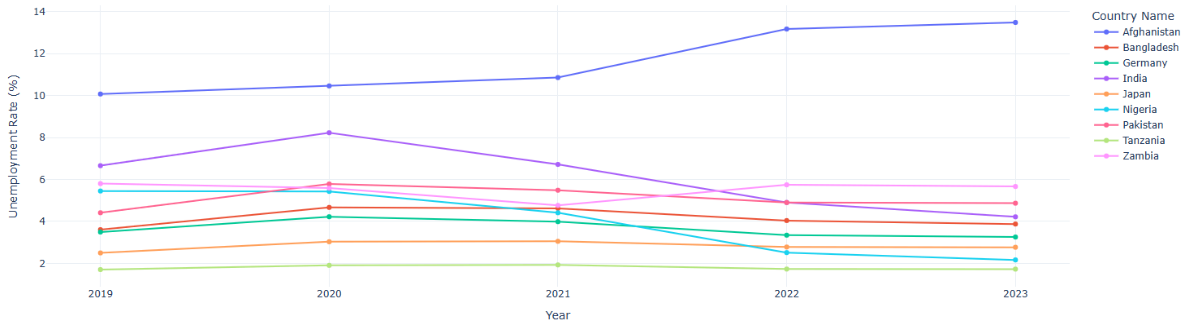

Figure 3. Male Unemployment Rate Trends by Country (2019 - 2023)

As observed in **Figure 3**, male unemployment trends from 2019 to 2023 show notable variations across countries. Afghanistan experienced a gradual rise from 10.07% in 2019 to 10.86% in 2020, followed by a sharp increase to 13.17% in 2022 and further to 13.48% in 2023. India showed an initial increase from 6.66% in 2019 to 8.23% in 2020 but then declined rapidly to 4.22% by 2023. Tanzania recorded the lowest male unemployment rate, reaching 1.73% in 2023, while Nigeria exhibited a significant drop from 4.41% in 2021 to 2.51% in 2022, making it the second-lowest in 2023.

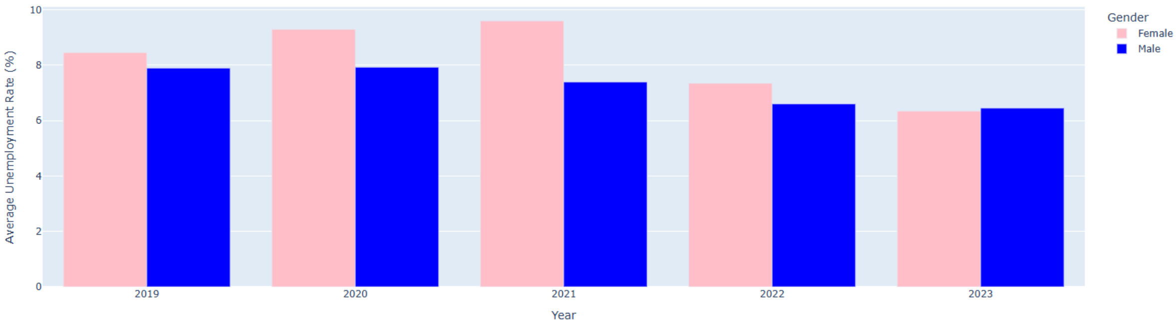

Figure 4.Global Unemployment Rate Trends by Gender (2019 - 2023)

Unemployment rates as in **Figure 4 s**how that female unemployment rose from 8.45% in 2019 to 9.6% in 2021, before declining to 7.34% in 2022 and 6.34% in 2023 often highlighting the uneven impact of economic challenges, with women frequently facing higher rates than men due to structural barriers, limited access to opportunities, and discriminatory practices. Interestingly, in 2023, the male unemployment rate of 6.44% slightly exceeded the female rate of 6.34%, highlighting a narrowing gender gap in unemployment.

Labor force participation further underscores these disparities, as cultural, social, and economic factors influence women's engagement in paid work, often leaving their potential untapped in many regions. The proportion of wage and salaried workers as in **Figure 5** is another critical indicator, showing differences in job security and earning stability. While men are more likely to hold formal, salaried positions, women are often overrepresented in informal and precarious employment, contributing to persistent wage gaps and economic vulnerability.

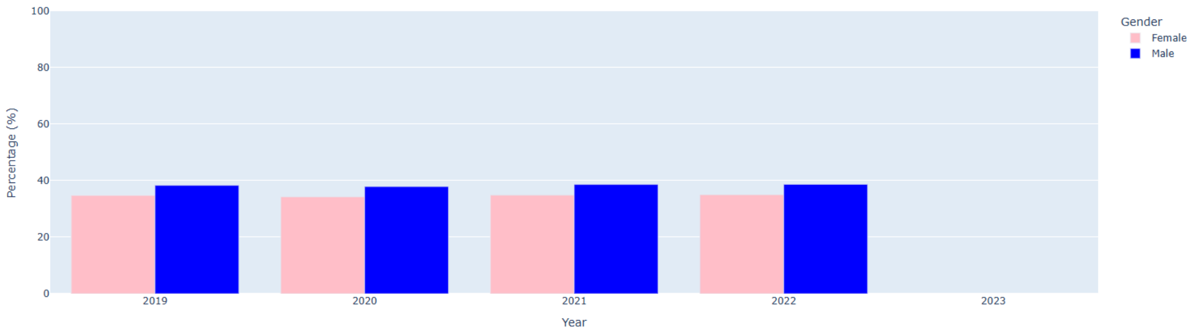

Figure 5. Wage Gap by Gender (2019 - 2023)

These indicators collectively reveal the multifaceted nature of economic inequalities and provide a lens through which we can assess progress and identify areas requiring targeted interventions to ensure equitable economic opportunities for all genders.

4.1.2 Educational Attainment

Primary and tertiary educational attainment for both females and males reveals significant gender disparities across various regions. In many countries, female enrollment rates in primary education are generally high, reflecting progress towards achieving gender parity in early education. **Figure 6 and Figure 7** show the variations in the enrolment rates of

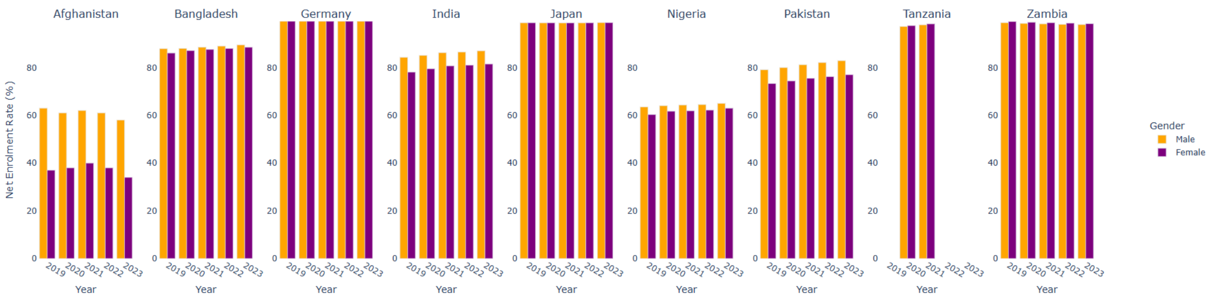

Figure 6. Net Enrollment Rate by Gender in Primary Education (2019 - 2023)

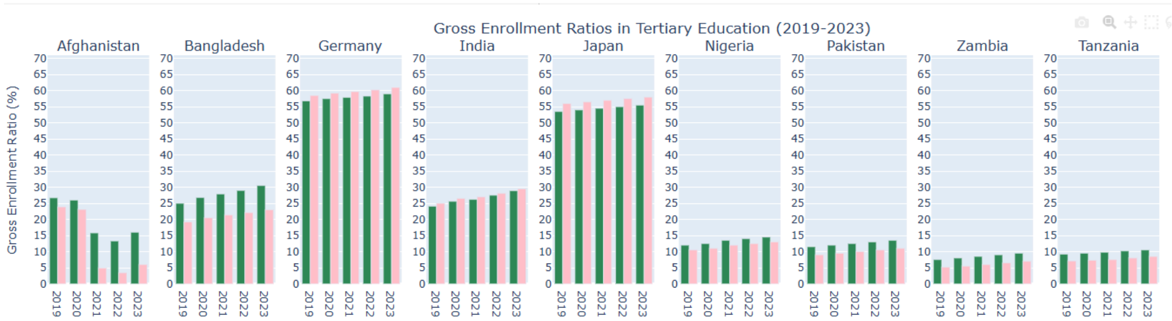

Figure 7. Gross Enrollment Ratios by Gender in Tertiary Education (2019 - 2023)

male and female in primary and tertiary education. Gender gaps often emerge at the tertiary level, where male enrollment tends to outpace female enrollment in certain regions due to factors like economic barriers, cultural norms, and societal expectations. Despite these challenges, the gap in tertiary education has been narrowing in many parts of the world, with increasing efforts to promote female participation in higher education. These trends underscore the importance of addressing barriers to education at all levels to ensure equal opportunities for both genders, ultimately empowering women and contributing to broader social and economic development.

4.1.3 Health and Survival

Health and survival outcomes often reflect significant gender disparities, with men and women facing distinct health challenges influenced by biological, social, and economic factors. In many regions, women tend to live longer than men, but they also face unique health risks, such as maternal mortality, reproductive health issues, and gender-based violence, which can impact their overall well-being. Men, on the other hand, generally experience higher mortality rates at younger ages due to factors like higher rates of violence, accidents, and lifestyle-related diseases. Additionally, access to healthcare services and health information is often influenced by gender norms and economic barriers, further exacerbating these inequalities.

As observed in **Figure 8**, Japan and Germany exhibit the highest life expectancy at birth. In Japan, females have a life expectancy of around 87 years, while males have a life expectancy of 84 years. In Germany, females' life expectancy ranges from 82 to 83 years, and for males, it is between 80 and 81 years. On the other hand, countries like Nigeria and

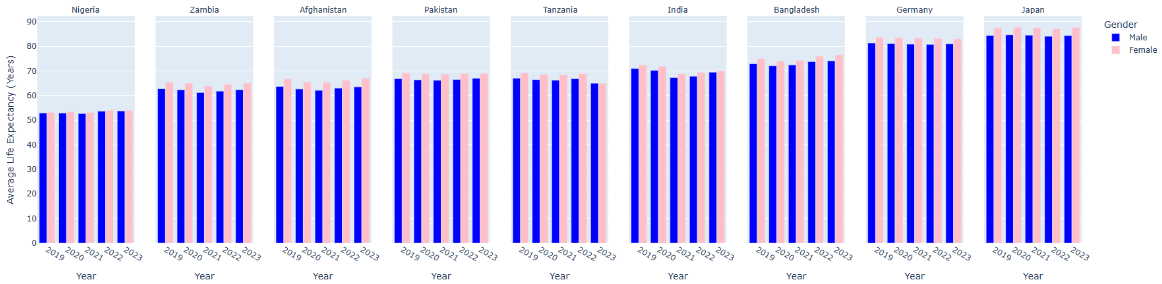

Figure 8. Average Life Expectancy at Birth by Gender and Country (2019-2023)

Zambia have much lower life expectancy rates. In Nigeria, females have a life expectancy of 53 years, while males range from 52 to 53 years. In Zambia, females' life expectancy is between 62 and 65 years, and males range from 61 to 62 years.

Addressing these gendered health disparities requires a focus on improving access to healthcare, promoting gender-sensitive health policies, and addressing the root causes of inequality in both health outcomes and access to care.

4.1.4 Political Empowerment

Political empowerment with respect to gender highlights the disparities in political representation and decision-making power between men and women. Despite progress in some regions, women remain underrepresented in national parliaments and political offices globally. Barriers such as gender norms, discrimination, and limited access to political networks hinder women's participation in politics. Efforts to promote gender equality in political representation are crucial for ensuring that women's voices are heard in shaping policies that impact their lives, contributing to more inclusive and equitable governance.

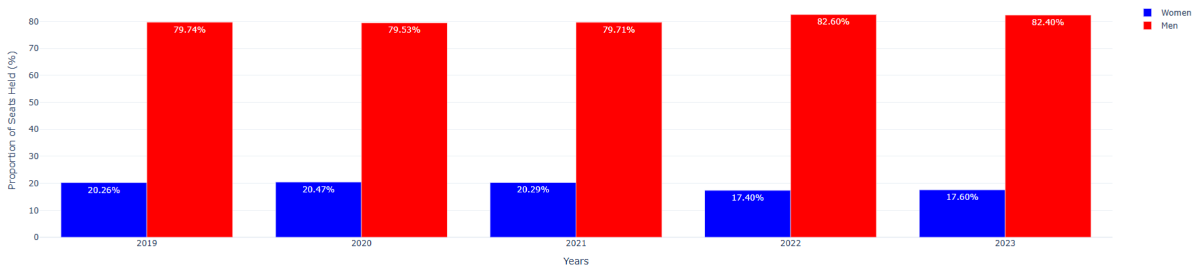

Figure 9. Proportion of seats held by Women and Men in National Parliament (2019 - 2023)

As observed in **Figure 9**, the proportion of men in politics consistently dominates compared to women across all years. Men's representation in political positions is nearly 82%, while women's representation ranges from 17% to 20%. This disparity underscores the ongoing gender gap in political empowerment and highlights the need for increased efforts to promote women's participation in political decision-making.

4.1.5 Online Accessibility

The female-to-male internet usage ratio in **Figure 10** highlights varying trends in gender equality across countries. Bangladesh shows significant improvement, driven by increased mobile internet use among women during the COVID-19 pandemic, despite cultural barriers.

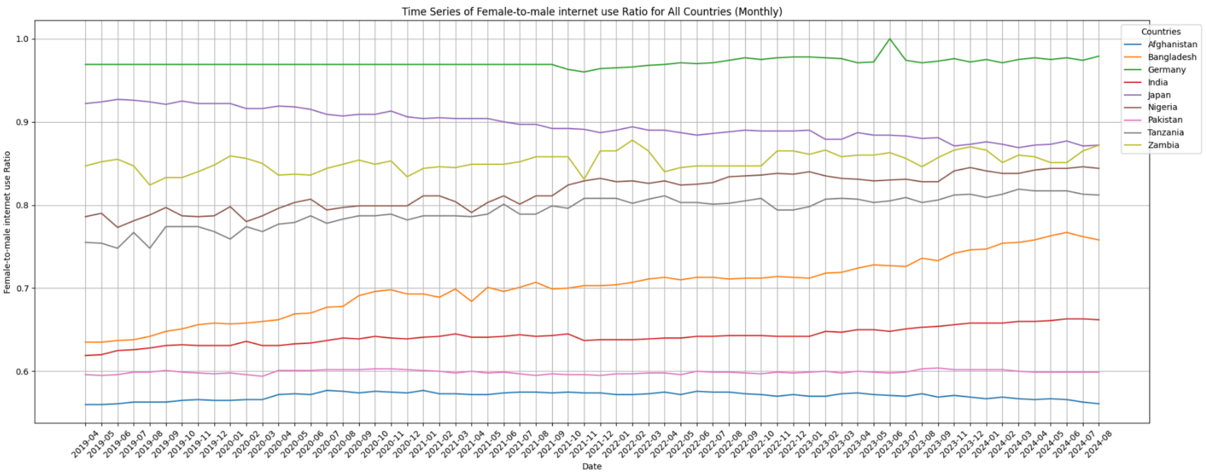

Figure 10. Female to Male Internet Usage Ratio (2019 - 2024)

Zambia experiences sharp fluctuations due to infrastructure issues and economic challenges, though rapid recoveries indicate quick resolutions. Japan exhibits a gradual decline, influenced by its aging population, traditional gender roles, and slower adoption of mobile internet among women. Germany shows a temporary spike in early 2023, likely due to a targeted digital campaign, followed by a return to typical patterns. These trends reflect diverse cultural, economic, and social factors shaping gender equality in internet access.

4.2 Child Labor EDA

The exploratory data analysis (EDA) for child labor focused on three primary aspects: trend analysis, correlation analysis, and clustering analysis. Each subsection provides a detailed examination of the data and its implications for understanding child labor dynamics across different countries.

4.2.1 Trend Analysis

Trend analysis examined temporal and sectoral patterns in child labor data. The study identified variations in child labor rates across countries and over time, emphasizing the influence of socio-political and economic factors.

Figure 11 provides a comprehensive overview of the child labor trend analysis categorized into three key aspects: economic and financial indicators, health and medical indicators, and food and disasters. This categorization allows for a focused investigation into how these dimensions influence child labor dynamics.

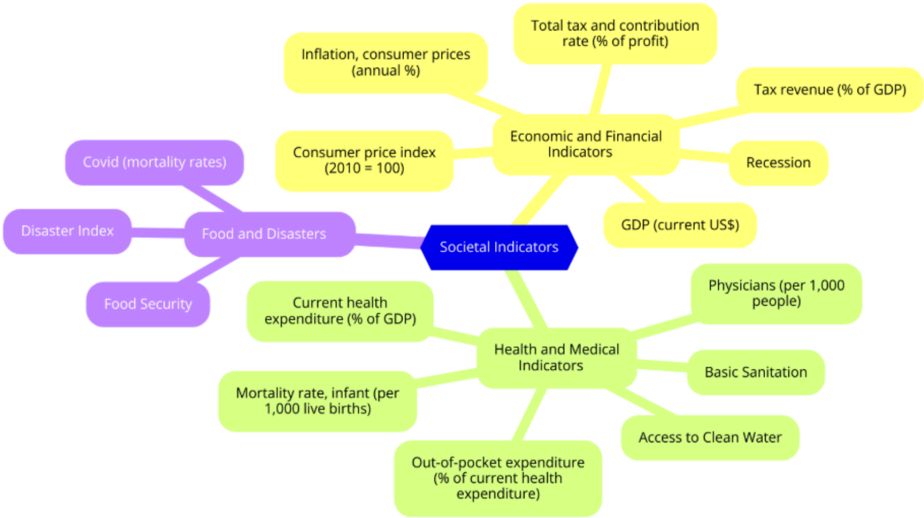

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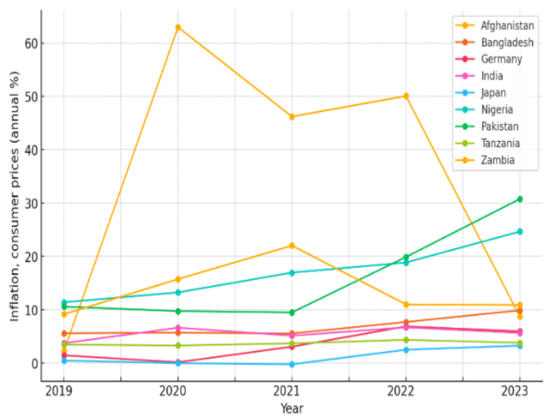
Figure 11. Trend Analysis in Three Aspects

Economic and financial indicators such as GDP, tax revenue, recession, inflation, and CPI were analyzed to understand their correlation with child labor rates. For instance, periods of high inflation or recession often coincide with increased child labor rates due to economic strain on households.



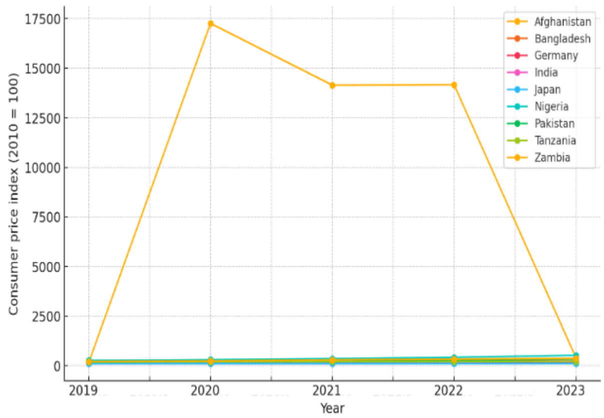
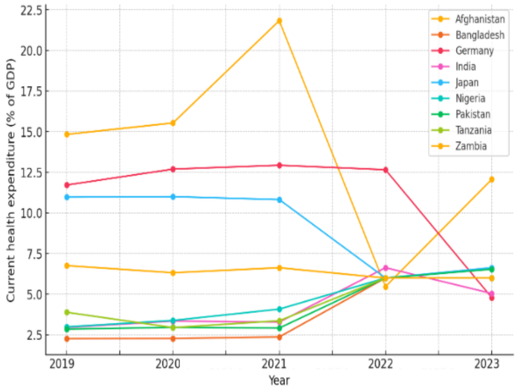


Figure 12. Trends in Inflation(Consumer Prices, Annual %) for Selected Countries (Left)
Figure 13. Trends in CPI (Consumer Price Index) for Selected Countries (Right)

Figure 12 shows the inflation trend between 2019 to 2023 in given countries, and **Figure 13** shows the CPI trend between 2019 to 2023 in given countries. It's clear that some countries, such as Afghanistan, are experiencing socio-political crises. This may lead to sharp increases in child labor rates, particularly during periods of economic instability and conflict.

As for current health expenditures shown in **Figure 14**, there is a huge decrease for Afghanistan in 2021, which may be related to its economic shocks, while other countries like Japan and India are generally stable between 2-8% of total GDP.



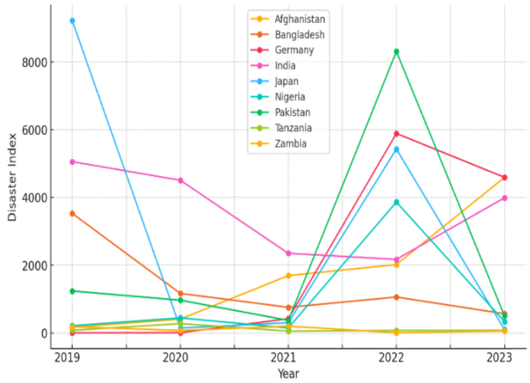


Figure 14. Trends in Health Expenditure in Total GDP(%) for Selected Countries (Left)
Figure 15. Trends in Disaster Index for Selected Countries (Right)

Figure 15 shows the trend of natural disaster index for given countries(COVID excluded). Many countries suffered from natural disasters between 2021 and 2023, which could be the cause of a decrease in economic and health indicators.

4.2.2 Correlation Analysis

Correlation analysis explored the relationships between child labor prevalence and socio-economic indicators, constructing a correlation matrix to quantify these associations.

To better understand the relationships between indicators and reduce redundancy, a correlation analysis was conducted, resulting in the creation of a heatmap. In the original dataset, over 50 variable pairs were identified as highly correlated, with a correlation coefficient exceeding 0.75. Examples of these pairs included Agricultural land (sq. km) and Land area (sq. km), as well as Birth rate and Fertility rate. After applying imputation techniques, only 11 pairs remained highly correlated using the same threshold.

The correlation heatmap shown in **Figure 16** revealed clusters of variables with high interdependence, which informed the exclusion of redundant indicators and streamlined the dataset.

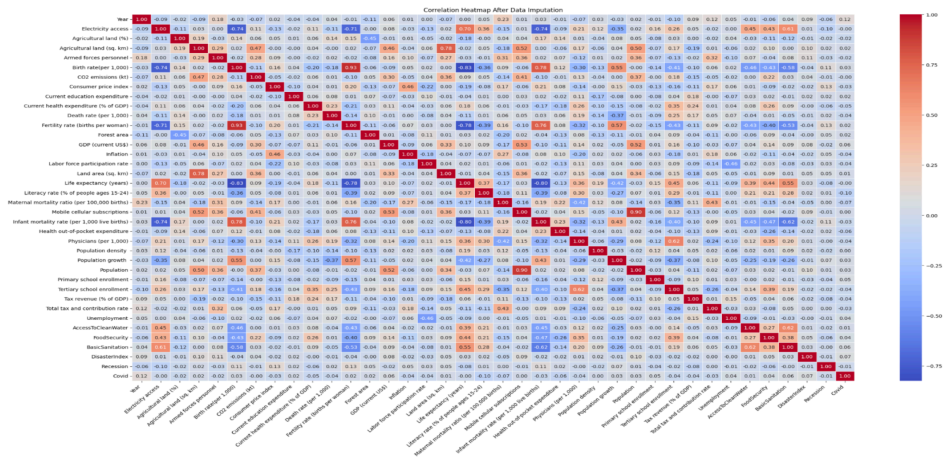


Figure 16. Figure Correlation Heatmap For Child Labor Data

Removing highly correlated variables reduced redundancy and improved the model's ability to isolate the impact of individual predictors on child labor. The indicators excluded due to high correlation included Fertility Rate, Life Expectancy, Infant Mortality, Mobile Cellular Subscriptions, Electricity Access, and Mean Years of Schooling. This feature reduction ensured a streamlined dataset while preserving the most representative variables for further analysis.

4.2.3 Clustering Analysis

Clustering analysis was performed to categorize countries based on shared socio-economic characteristics related to child labor. This approach aimed to identify distinct groupings that could provide insights into regional patterns and underlying factors contributing to child labor.

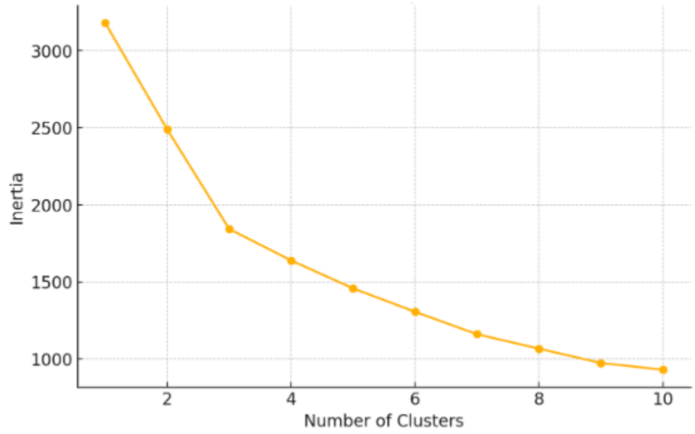


Figure 17. Elbow Method for Optimal Clusters

Using the elbow method in **Figure 17**, three clusters were determined as the optimal number for segmentation.

One example of our clustering is provided in **Figure 18**, which clusters countries based on disaster index and food security volatility. In the top left of the scatter plot, Columbia is notable for experiencing the highest food security volatility despite facing relatively few environmental disasters due to prolonged internal conflict and the negative impact of climate change, including droughts and rising sea levels. Other selected countries are also shown in the figure, with Japan and India suffering most from natural disasters while keeping food security steady.

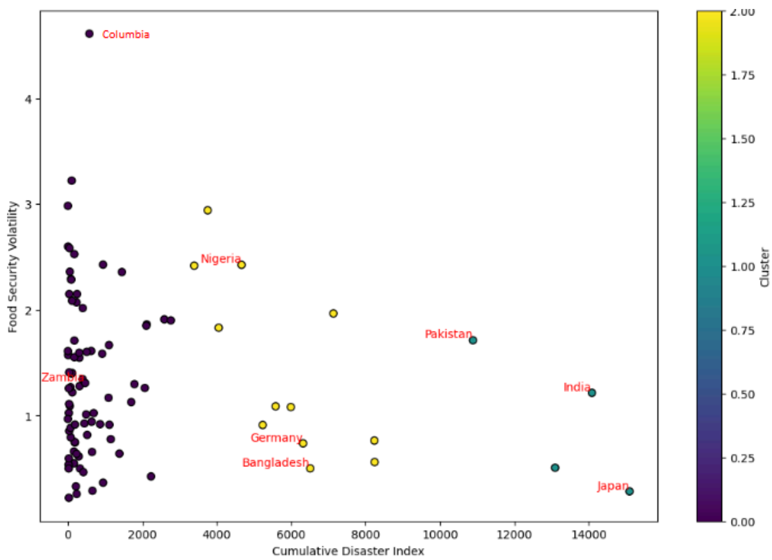


Figure 18. Clustering Analysis on Disaster Index and Food Security Volatility

The clustering results did not reveal distinct or meaningful groupings, reflecting the inherent complexity and overlap of socio-economic characteristics across countries.

Nonetheless, the analysis provided valuable insights into the underlying data structure and contributed to shaping subsequent modeling approaches.

5. Model Development

5.1 GGI Model Methodology and Model Development

5.1.1 Model Introduction

The Global Gender Gap Index (GGI) prediction model is designed to quantify gender disparities across critical dimensions, including economic participation, educational attainment, health and survival, and political empowerment. By adopting a data-driven approach, the model aims to uncover socio-economic factors contributing to these disparities and provide actionable insights to guide policy interventions.

To construct the GGI prediction model, three regression techniques were applied: Lasso, Ridge, and Elastic Net. Among these, Elastic Net emerged as the most effective method due to its ability to manage multicollinearity while maintaining a balance between model complexity and predictive power. Elastic Net combines the strengths of Lasso and Ridge regression, facilitating efficient feature selection and coefficient shrinkage, which optimizes both predictive accuracy and interpretability.

The primary objective of the GGI model is to identify the key factors driving gender disparities and to provide a robust framework for monitoring progress in gender equality initiatives. The model incorporates data from reputable sources such as the World Bank and International Labour Organization (ILO), as well as supplementary metrics like the Facebook Gender Gap Index (FBGGI). By integrating these datasets, the GGI model captures both offline and online dimensions of gender inequality, offering a comprehensive perspective on this critical issue.

The techniques employed in the GGI model, such as Elastic Net, share methodological similarities with those used in the child labor model, which will be elaborated on in Section 5.2.1.

5.1.2 Hyperparameter Tuning

To maximize the performance and generalizability of the Elastic Net model, we employed two complementary hyperparameter tuning strategies: Grid Search and Leave-One-Out Cross-Validation (LOOCV).

Grid Search was used to systematically explore combinations of hyperparameter values for optimal model performance. Specifically, we tuned the Elastic Net parameters alpha (regularization strength) and the L1 ratio (balance between Lasso and Ridge penalties). The search space included a range of values for alpha and L1 ratio, where alpha controlled the penalty's overall strength, and the L1 ratio determined the mix of Lasso (L1) and Ridge (L2) regularization. The optimized hyperparameters, Alpha: 0.00072 and L1 Ratio: 0.98, strike a balance between feature selection and coefficient stability, ensuring the model remains both interpretable and accurate. These values ensured the model achieved a balance between selecting relevant features (via Lasso) and stabilizing coefficient estimates (via Ridge).

Beyond tuning regression parameters, Grid Search was also applied to optimize the weights assigned to each socio-economic indicator in the model. The indicators, categorized into dimensions such as education, employment, technology access, health, and politics, were assigned weight ranges based on prior research and domain expertise.

After rigorous testing, the final weights were determined as follows: FBGGI has the highest weight of all, which is 0.34. Primary and Secondary Education received a weight of 0.15 each, Tertiary Education was assigned 0.13, while indicators like Global Unemployment, Internet Access, and Mobile Access were weighted at 0.07, 0.1, and 0.1, respectively. Lesser weights were allocated to Health and Politics, each receiving 0.05. These optimized weights reflected the varying influence of the indicators on GGI.

To validate the robustness of these parameters and weights, LOOCV was employed. This method involved iteratively training the model on all but one data point and testing on the excluded sample, thereby ensuring that the model's predictions generalized well across different datasets. By minimizing overfitting and enhancing reliability, LOOCV provided a strong foundation for the model's predictive framework.

5.1.3 Model Evaluation

Models were trained using 2023 data to predict 2022 GGI values, with Elastic Net demonstrating superior predictive performance. **Figure 19** illustrates model predictions for 2022, highlighting Elastic Net's superior performance in accurately estimating GGI across diverse socio-economic contexts. Highlighted cells in green means that the model's prediction on that country is the closest to the actual value.

	country	2022	Predicted_2022_Ridge	Predicted_2022_Lasso	Predicted_2022_ElasticNet
0	Afghanistan	0.155040	0.176824	0.606409	0.167865
1	Bangladesh	0.404000	0.420162	0.543679	0.424270
2	Germany	1.047000	1.002014	0.503407	1.012453
3	India	0.314829	0.321410	0.544792	0.316180
4	Japan	0.755227	0.804517	0.585134	0.780530
5	Nigeria	0.700360	0.710813	0.582204	0.699492
6	Pakistan	0.196900	0.223352	0.578967	0.212428
7	Tanzania	0.600400	0.612416	0.569516	0.615974
8	Zambia	0.900500	0.814739	0.562499	0.826357

Figure 19. Predictions for all Three Models on 2022 Based on 2023 Data

The performance of all three models were evaluated using two key metrics: RMSE and R^2. Root Mean Square Error (RMSE) measures the average magnitude of prediction errors in a model. A lower RMSE indicates that the model's predictions are closer to the actual observed values, reflecting high precision. R-Squared (R^2) quantifies the proportion of variance in the dependent variable that can be explained by the model's predictors. Higher R^2 values signify that the model effectively captures the relationships between the independent variables and the GGI.

Model (2022)	RMSE	R^2
Elastic Net	0.0251	93.78%
Ridge	0.0367	86.72%
Lasso	0.1006	0.00%

Figure 20. RMSE and R^2 of all three models

Figure 20 compares the RMSE and R^2 metrics across models, with Elastic Net achieving the lowest RMSE and highest R^2. These results underscore the Elastic Net model's capability to identify and represent the underlying socio-economic dynamics influencing gender disparities.

Elastic Net was particularly advantageous for this application due to the high interdependency among socio-economic variables. Its regularization mechanisms allowed the model to balance the trade-off between underfitting and overfitting, ensuring robust and generalizable predictions.

5.2 Child Labor Model Methodology and Model Development

5.2.1 Model Selection

The development of the Child Labor Model initially included the Facebook Gender Gap Index (FBGGI) as a feature. However, due to the limited availability and size of the FBGGI dataset, the model exhibited constant overfitting. To address this limitation, the FBGGI variable was excluded from subsequent modeling efforts. Various modeling techniques were explored to identify the most effective approach for predicting child labor indicators. The following methodologies were employed:

Lasso Regression: Lasso (Least Absolute Shrinkage and Selection Operator) Regression is a linear regression technique that incorporates an L1 penalty to the loss function. This penalty shrinks some coefficients to zero, effectively performing feature selection. It is particularly effective for datasets with a large number of features, simplifying the model by managing multicollinearity and reducing overfitting.

$$\text{Loss} = \text{RSS} + \lambda \sum_{j=1}^{p} |\beta_j|$$

Ridge Regression: Ridge Regression applies an L2 penalty to the loss function, shrinking coefficients while retaining all features in the model. This technique is suitable for datasets with multicollinearity among predictors, as it prevents overfitting by pulling coefficients closer to zero.

$$\text{Loss} = \text{RSS} + \lambda \sum_{j=1}^{p} \beta_j^2$$

ElasticNet Regression: ElasticNet Regression combines the L1 (Lasso) and L2 (Ridge) penalties, balancing feature selection and coefficient shrinkage. This model is useful for high-dimensional datasets with correlated predictors, combining the strengths of Lasso's feature selection and Ridge's multicollinearity handling.

$$\text{Loss} = \text{RSS} + \lambda_1 \sum_{j=1}^{p} |\beta_j| + \lambda_2 \sum_{j=1}^{p} \beta_j^2$$

ElasticNetCV with Gradient Boosting Regressor: ElasticNetCV incorporates automatic hyperparameter tuning through cross-validation, while Gradient Boosting builds ensemble models sequentially to correct errors from prior models. This approach is well-suited for capturing nonlinear relationships and complex interactions, combining ElasticNet's feature selection with Gradient Boosting's ensemble learning.

H2O AutoML: H2O AutoML automates the training and evaluation of multiple models, optimizing hyperparameters to identify the best-performing algorithm. This framework is effective for exploratory analysis and model comparison, as it systematically tests various algorithms to minimize Mean Squared Error (MSE).

$$F_m(x) = F_{m-1}(x) + h_m(x)$$

Random Forest: Random Forest is an ensemble learning method that constructs multiple decision trees during training and averages their predictions. This approach improves accuracy and prevents overfitting.

$$\hat{y} = \frac{1}{N}\sum_{i=1}^{N}\hat{y}_i$$

Hyperparameter Tuning Approaches:

1. **Grid Search:** Systematically explores hyperparameter combinations to optimize model performance.
2. **Random Search:** Samples hyperparameter values randomly from predefined distributions for optimization.

Model Comparison and Final Selection:

Extensive testing of various models revealed distinct strengths and weaknesses. The results are summarized as follows:

Model	RMSE	R²
Random Forest (max_depth=10, min_samples_leaf=4, min_samples_split=10)	0.0785	48.70%
Ridge (alpha=0.3196, max_iter=1000)	0.0730	56.70%
Lasso (alpha=0.000411)	0.0716	58.40%
Elastic Net CV (Gradient Boosting Regressor)	0.0639	26.98%
Elastic Net (alpha=0.01, l1_ratio=0.9)	0.0671	53.69%
Random Forest (Grid Search)	0.0863	45.30%
Random Forest (Random Search)	0.0868	44.70%
H2O (Auto ML)	0.0649	34.12%

Figure 21. Child Labor Model Validation and Comparison

1. **Lasso Regression:** Achieved the highest R^2 value (58.40%), demonstrating strong explanatory power. However, its relatively higher RMSE (0.0716) indicated reduced prediction accuracy.
2. **ElasticNet with Gradient Boosting:** Recorded the lowest RMSE (0.0639), showcasing excellent prediction accuracy but a very low R^2 value (26.98%), suggesting potential overfitting and limited generalizability.
3. **ElasticNet Regression (Final Selection):** ElasticNet with hyperparameters α=0.01 and l1_ratio=0.9 emerged as the most balanced model, achieving a low RMSE (0.0707) and a high R^2 value (58.37%). This model effectively balanced prediction accuracy and explanatory power.

As shown in **Figure 21**, the ElasticNet model with α=0.01 and l1_ratio=0.9 was selected as the final model for its robust and interpretable predictions. It combines low error rates with strong variance explanations, aligning with the project's goals of accurately predicting child labor indicators while providing insights into underlying drivers. By adopting ElasticNet, the analysis ensures reliable predictions and meaningful interpretations to guide future policy and intervention strategies.

5.2.2 Feature Importance

To understand how the indicators affect child labor, statistical analyses were conducted. The ElasticNet model with assigned weights was utilized to minimize MSE through weight grid search, and the indicators were ranked based on their significance. Rankings were also derived from feature importance metrics using ElasticNet and Random Forest models. Key indicators were categorized based on their overall significance and consistency across ranking methods:

1. **Critical Indicators:** These factors consistently ranked among the most significant across multiple evaluation methods, indicating their substantial impact on child labor. Examples include the Human Development Index, Basic Sanitation, and Birth Rate (per 1,000).
2. **Significant Indicators:** These factors demonstrated strong importance in at least one ranking method and were frequently highlighted in the analysis. Examples include Total Tax and Contribution Rate, Literacy Rate (% of people ages 15–24), Tertiary School Enrollment, Recession, Health Out-of-Pocket Expenditure, Physicians (per 1,000), and Unemployment.
3. **Additional Influential Factors:** Other factors, such as Population, GDP (current US$), Death Rate (per 1,000), Armed Forces Personnel, Maternal Mortality Ratio (per 100,000 births), Labor Force Participation Rate, Disaster Index, CO2 Emissions (kt), Access to Clean Water, Consumer Price Index, and Agricultural Land (sq. km) were also identified as contributing to child labor to varying degrees.

Weight	ELN importance	RF importance
Human Development Index (value)	Labor force participation rate	**Human Development Index (value)**
BasicSanitation	**Birth rate(per 1,000)**	Maternal mortality ratio (per 100,000 births)
Total tax and contribution rate	Armed forces personnel	**Birth rate(per 1,000)**
Literacy rate (% of people ages 15-24)	DisasterIndex	Physicians (per 1,000)
Tertiary school enrollment	CO2 emissions (kt)	Unemployment
Birth rate(per 1,000)	Maternal mortality ratio (per 100,000 births)	Tertiary school enrollment
Recession	**BasicSanitation**	Death rate (per 1,000)
Health out-of-pocket expenditure	AccessToCleanWater	Labor force participation rate
Physicians (per 1,000)	Consumer price index	Forest area
Unemployment	**Human Development Index (value)**	Total tax and contribution rate
Population	Agricultural land (sq. km)	Tax revenue (% of GDP)
GDP (current US$)	Population	**BasicSanitation**
Death rate (per 1,000)	Forest area	Inflation
Armed forces personnel	Inflation	Consumer price index
Covid	Land area (sq. km)	Recession

Figure 22. Feature Importance Across Weight, Elastic Net, and Random Forest Models with Highlights (Red: Common to All Columns, Yellow: Present in Two Columns)

This categorization serves as a reliable foundation for determining impactful features. By focusing on these indicators, the analysis ensures a targeted approach to understanding and addressing the drivers of child labor effectively. As illustrated in **Figure 22**, these rankings and categorizations summarize feature importance across Weight, Elastic Net, and Random Forest models.

5.2.3 Key Indicators

The identification of key indicators was a pivotal part of the analysis, focusing on features with significant predictive power and practical relevance. These indicators were selected based on their frequency of usage in the ElasticNet model during leave-one-out analysis (LOOCV) and their overall contribution to the model's performance.

Top 10 Key Indicators:

The following indicators were identified as the most impactful:

1. **Human Development Index (HDI):** A composite measure reflecting a country's social and economic development, HDI was the most influential factor in determining child labor rates.
2. **Maternal Mortality Ratio (per 100,000 births):** This variable reflects healthcare quality and indirectly correlates with economic and social conditions that influence child labor.
3. **Birth Rate (per 1,000):** High birth rates are associated with increased economic pressure on families, leading to higher child labor rates.
4. **Labor Force Participation Rate:** The proportion of the working-age population that is economically active directly affects child labor dynamics.
5. **Physicians (per 1,000):** The availability of healthcare services indicates broader developmental factors impacting child welfare.
6. **Basic Sanitation:** Access to sanitation is closely linked to living conditions, which have a significant influence on child labor.
7. **Armed Forces Personnel:** This indicator reflects national stability, with periods of conflict often exacerbating child labor issues.
8. **Death Rate (per 1,000):** High death rates signal poor living conditions, which are linked to increased child labor.
9. **Tertiary School Enrollment:** Higher enrollment rates in tertiary education indicate improved educational infrastructure, reducing the prevalence of child labor.
10. **Unemployment:** Elevated unemployment rates can increase reliance on child labor as families seek additional income sources.

Insights from Feature Selection:

The feature selection process underscored the importance of these indicators in understanding child labor. Frequent selection of these features in the ElasticNet model highlighted their critical role in shaping child labor trends. By focusing on these key indicators, the analysis provides actionable insights to guide policy and intervention strategies, ensuring a targeted and effective approach to addressing child labor challenges.

6. Performance and Results

6.1 GGI Model Results

The performance and results of the selected GGI model demonstrate the effectiveness of the Elastic Net model in capturing the socio-economic dynamics of gender disparities. With an R^2 of 93.78%, the model explains nearly all the variation in the Gender Gap Index (GGI) based on the input features, indicating strong predictive power. The low RMSE of 0.0251 shows that the model's predictions are very close to the actual GGI values. Elastic Net emerged as the most effective model for predicting GGI values for both 2022 and 2023 using the leave-one-out method. It consistently delivered highly accurate predictions, capturing the complex relationships between socio-economic factors and gender disparities. The model's ability to manage overlapping influences among variables allowed it to balance simplicity and complexity, resulting in robust and reliable predictions for many countries, as previously illustrated in **Figure 19**.

6.2 Child Labor Model Results

The performance of the child labor model also highlighted Elastic Net as the best-performing model, offering the most balanced results between R^2 and RMSE. Incorporating the Facebook Gender Gap Index (FBGGI) initially yielded promising results, with an RMSE of 0.013 and an R^2 of 97.23%. However, since only six observations had both FBGGI and child labor data, overfitting likely occurred, leading to the exclusion of this approach.

When comparing models on the decimal-scaled data, an RMSE of 0.0707 and an R^2 of 58.37% were observed. After applying the Yeo-Johnson transformation to handle negative values, the R^2 increased slightly to 59.18%. When RMSE was scaled back to the original units, it translated to 7.2543 percentage points, indicating that the model's predictions were off by an average of 7.25 percentage points. Given the average child labor rate of 14.36%, this corresponds to an RMSE percentage of the mean at 50.50%, signaling room for improvement. The power transformation also adjusted the model's optimal hyperparameters, with alpha at 0.19 and an l1 ratio of 0.4.

Figure 23 shows the predicted vs. actual child labor values, revealing that the model performs reasonably well overall but struggles with extreme values. It tends to overpredict low child labor percentages while underpredicting very high percentages, contributing to the higher RMSE.

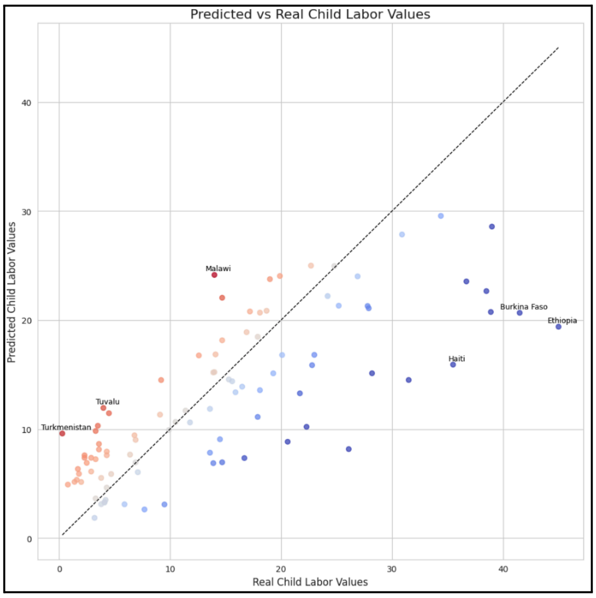


Figure 23. Predicted vs Real Child Labor Values

To address these challenges, clustering based on the Human Development Index (HDI), the strongest predictor, and maternal mortality ratio was explored. Using several approaches, agglomerative clustering with four clusters produced the best results. Agglomerative clustering, a hierarchical method, starts with each observation as its own cluster and iteratively merges the closest clusters based on a defined distance metric until the desired number of clusters is reached (Datanovia). While clustering showed potential, the results were mixed: HDI-based clusters resulted in an R^2 of 0.599 and RMSE of 7.44, while combining HDI and maternal mortality ratio slightly shifted the metrics to an R^2 of 0.56 and RMSE of 7.18. These trade-offs highlight the need for further refinement to improve model performance.

6.3 Correlation Analysis

The correlation analysis began with the creation of a heatmap to visualize the correlation matrix, which was divided into two sections: one for features with higher correlation and another for features with lower correlation. The y-axis represents the indicators used to predict child labor, while the x-axis represents the indicators used to predict the Gender Gap Index (GGI).The initiall focus was placed on the matrix with higher correlations, as it highlights potential indicators that may be useful in analyzing the other dataset, as shown in **Figure 24**.

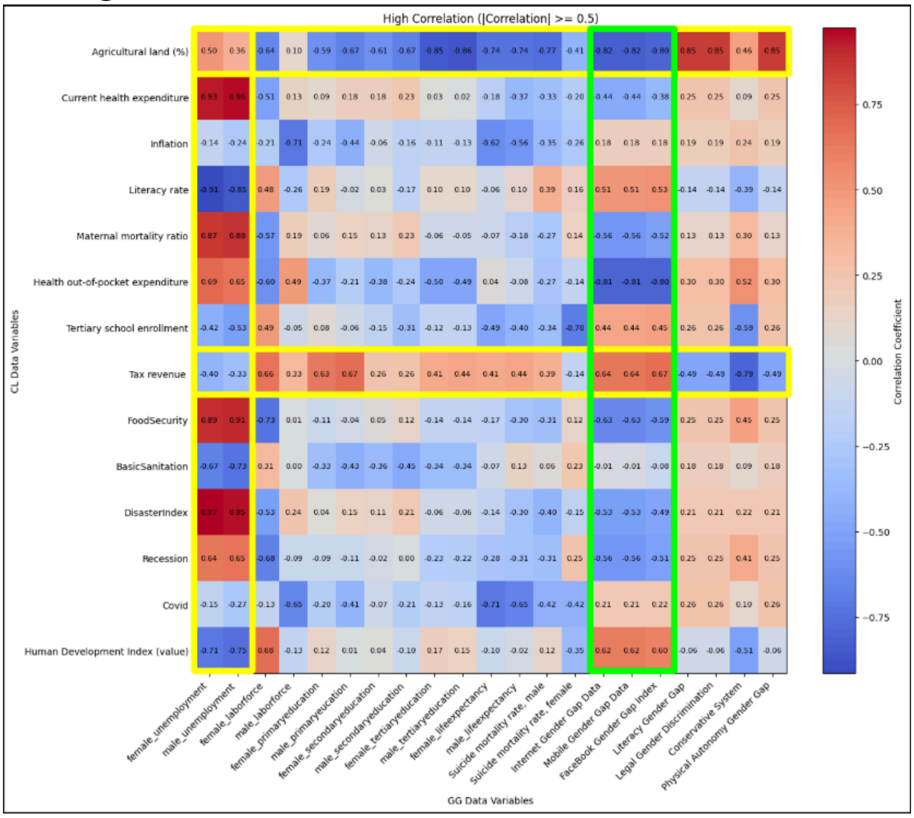


Figure 24. Heatmap of High Correlation Indicators from Child Labor and GGI Data

Within the highly correlated matrix, several indicators were identified as key relationships between child labor and GGI. For child labor, agricultural land percentage, tax revenue, and both female and male unemployment rates in GGI were found to have significant correlations, suggesting they may be useful for further analysis. In the green box, three additional indicators, including online accessibility and the Facebook Gender Gap Index (FBGGI), were also noted as highly correlated with GGI. These insights suggest that these variables may provide valuable input when modeling child labor and GGI.

Following the correlation analysis, countries were categorized based on their development levels, and a scatter plot was created to compare predicted child labor rates with GGI scores. In highly developed countries, a clear trend emerged showing that child labor tends to be lowest when GGI approaches 0.7 (**Figure 25**).

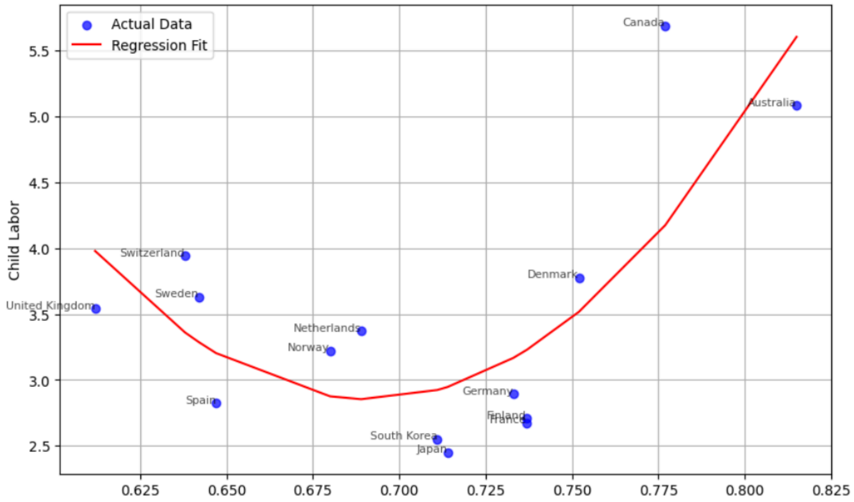


Figure 25. Scatter Plot of High-Development Countries' Child Labor and GGI Values with Quadratic Regression Model

The corresponding RMSE for this analysis was 0.5879, and the R^2 value was 58.58%, indicating a moderate level of predictive accuracy. However, for countries with lower development levels, no clear correlation between child labor and GGI was observed. Notably, three countries—Mozambique, Kenya, and Namibia—stood out due to their economic crises, which may have contributed to the rise in child labor rates as shown in **Figure 26**.

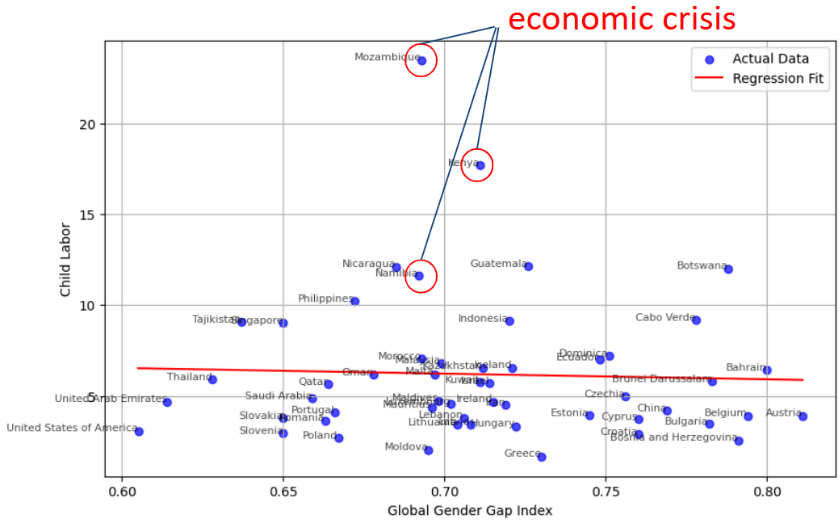


Figure 26. Scatter Plot of Low-Development Countries' Child Labor and GGI Values with Linear Regression Model

In conclusion, while the analysis reveals strong correlations in more developed nations, further exploration is needed to better understand the complexities of child labor and gender gap dynamics in lower-income countries, where external factors such as economic instability may influence the outcomes.

6.4 Economic Cost Saving

The economic costs of the project were evaluated by comparing the expenses of hiring a team in Rochester, NY, to the projected costs of conducting on-site research in multiple countries. The cost of hiring three data scientists and two data analysts for two months (working 10 hours per week) in Rochester amounted to approximately $20,011.20, which includes salaries based on the average hourly wage for data scientists ($58.22) and data analysts ($37.74).

In contrast, conducting on-site research in multiple countries would significantly increase costs, factoring in travel, accommodation, security measures, and other logistical expenses. For instance, on-site research in Afghanistan alone would total approximately $80,407.33, including round-trip flights, bodyguard fees, accommodation, and additional costs for translators, guides, and transportation. Other countries, such as Germany, India, and Nigeria, also show similarly high costs, ranging from $7,006 to $127,932, depending on the specific country and security needs.

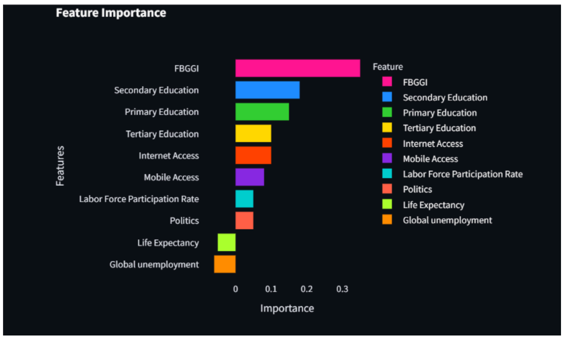
By opting for the remote approach using the team in Rochester, a total savings of $24,569,239 is achieved, surpassing the original on-site research expenses. This translates into savings of over 24 million dollars, along with the equivalent of 70.7 years of work, showcasing the substantial cost-efficiency of the project. Therefore, the decision to hire a team remotely in Rochester not only minimizes financial outlays but also avoids the complex logistical challenges and security risks associated with on-site international research.

6.5 Dashboard Development

The development of the GGI (Global Gender Gap Index) Dashboard and Child Labor Dashboard was guided by the critical need to analyze, interpret, and present complex datasets in an accessible and visually engaging manner. Built using libraries like Streamlit for interactivity, Pandas and NumPy for data manipulation, and Matplotlib and Plotly for visualization, these dashboards were designed to provide actionable insights into pressing global issues like gender disparities and child labor trends. Through data-driven storytelling, they reveal patterns, highlight disparities, and explore key contributing factors, making them valuable tools for understanding and addressing these challenges.

The GGI Dashboard focused on tracking and analyzing gender gap indicators such as labor force participation, education levels, internet and mobile access, and parliamentary representation. We incorporated advanced visuals like line charts, treemaps, and scatter plots to identify trends and correlations over time. An interactive map further enabled users to explore regional disparities. Complementing this, the GGI Model Dashboard employed predictive analytics to showcase how various indicators contributed to gender disparities, offering insights through feature importance plots, residuals analysis, and predictions vs. actual comparisons. This dual approach of descriptive and predictive analysis made the GGI dashboards indispensable for both understanding historical trends and forecasting future scenarios.

Similarly, the Child Labor Dashboard was built to analyze trends in child labor across regions, focusing on the interplay of economic, educational, and environmental indicators. Using comprehensive visuals, we identified critical factors influencing child labor rates, presenting insights through dynamic maps, comparative bar charts, and correlation analyses. The Child Labor Model Dashboard extended this analysis by leveraging machine learning models to predict and explain the impact of these factors, enhancing the utility of the dashboard for policy-making and advocacy.



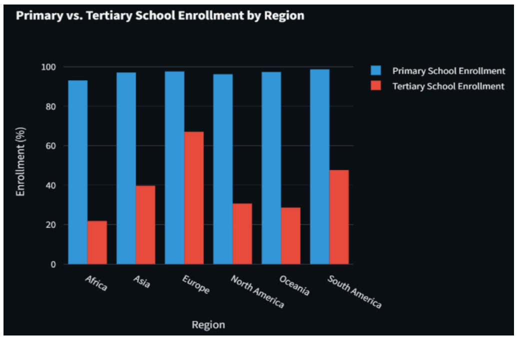


Figure 27. Feature Importance for GGI Model (Left)
Figure 28. Primary VS. Tertiary Education Enrollment by Region (Right)

Figures 27 and **28** showcase a concise overview of our interactive dashboard. These figures provide a glimpse into the tool's functionality, emphasizing its ability to visualize data insights effectively.

7. Conclusion and Future Works

This analysis uncovered distinct and overlapping factors influencing child labor and gender gaps in education and employment. Cross-sectional analyses revealed that child labor is significantly associated with levels of economic development, alongside a range of social, cultural, and political factors. For instance, countries with lower economic development and weaker labor protections tend to exhibit higher rates of child labor. On the other hand, tertiary education enrollment is strongly linked to gender disparities, highlighting systemic challenges that hinder women's access to higher education. The correlation analysis also uncovered overlapping factors that merit further investigation. A particularly interesting finding emerged regarding the relationship between tertiary school enrollment, household chores, and child labor. This dynamic varies significantly between regions such as Africa and Europe. In Africa, high child labor rates and low tertiary enrollment are often driven by children's participation in household responsibilities, disproportionately affecting girls. In contrast, European countries display lower child labor rates and higher tertiary enrollment due to stronger educational and labor policies. While current data limitations prevent robust predictive modeling through machine learning, these patterns suggest areas where targeted interventions could make a meaningful impact. Expanding the scope of data collection to include more granular insights on household chores and their impact on education and labor outcomes is critical for advancing this understanding.

To deepen these insights, future efforts should focus on collecting and analyzing more comprehensive datasets, particularly longitudinal data, to understand how these issues evolve over time. The current reliance on time-series data collected once every five to ten years limits the ability to track short-term trends or predict emerging risks. Developing more frequent and detailed datasets would enable researchers to create dynamic dashboards to monitor the risk of child labor in real time, especially in relation to economic development. Such tools would provide policymakers with actionable insights to design timely interventions. Additionally, a deeper exploration of the relationship between household chores, tertiary education enrollment, and child labor is warranted. Regional comparisons, such as those between Africa and Europe, illustrate the importance of examining these dynamics through a contextual lens. For example, countries in Africa could benefit from policies that alleviate household burdens on children, enabling them to pursue education, particularly at the tertiary level. Moreover, future research should explore the social and cultural drivers of these disparities and aim to identify actionable strategies to reduce them. Strengthening time-series data, investigating household labor patterns, and expanding the scope of analysis to incorporate economic, social, and cultural dimensions will be key steps toward addressing child labor and promoting gender equity on a global scale.

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